

Week 6

Keel Bone Chicken Project

02/09/26



College of Engineering

This Week

Weekly Progress

- Furthered Data Analysis of IMU
 - New Study Conducted
 - Verification Successful
- Research on RF to DC Converters
 - 915 MHz Band Research Paper
- DigiKey Parts Ordering
 - Parts List Curated – F Number Required (Procurement Office)
- Agriculture Components
 - Sent electrical components needed for study to department

Furthered Data Analysis of IMU

New Study Conducted

- Ran study for longer period of time
~60 minutes
- Results were very similar (next slide)
- Data analysis was done with the following Python script

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

def analyze_z_direction(file_path):
    try:
        df = pd.read_csv(file_path)
    except FileNotFoundError:
        print(f"Error: File '{file_path}' not found.")
        return

    df['timestamp'] = pd.to_datetime(df['timestamp'])
    df = df.sort_values('timestamp').reset_index(drop=True)

    lsb_to_g = 1 / 4096.0
    g_to_ms2 = 9.80665

    df['Az_ms2'] = df['accelerometer_Az'] * lsb_to_g * g_to_ms2

    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 8), sharex=True)

    ax1.plot(df['timestamp'], df['Az_ms2'], color='b', linewidth=1.0)
    ax1.set_ylabel('Acceleration Z ($m/s^2$)', fontsize=12)
    ax1.set_title('Z-Axis Acceleration', fontsize=14)
    ax1.grid(True, which='both', linestyle='--', alpha=0.7)

    ax2.plot(df['timestamp'], df['gyroscope_z'], color='g', linewidth=1.0)
    ax2.set_ylabel('Gyroscope Z (LSB)', fontsize=12)
    ax2.set_title('Z-Axis Gyroscope', fontsize=14)
    ax2.grid(True, which='both', linestyle='--', alpha=0.7)

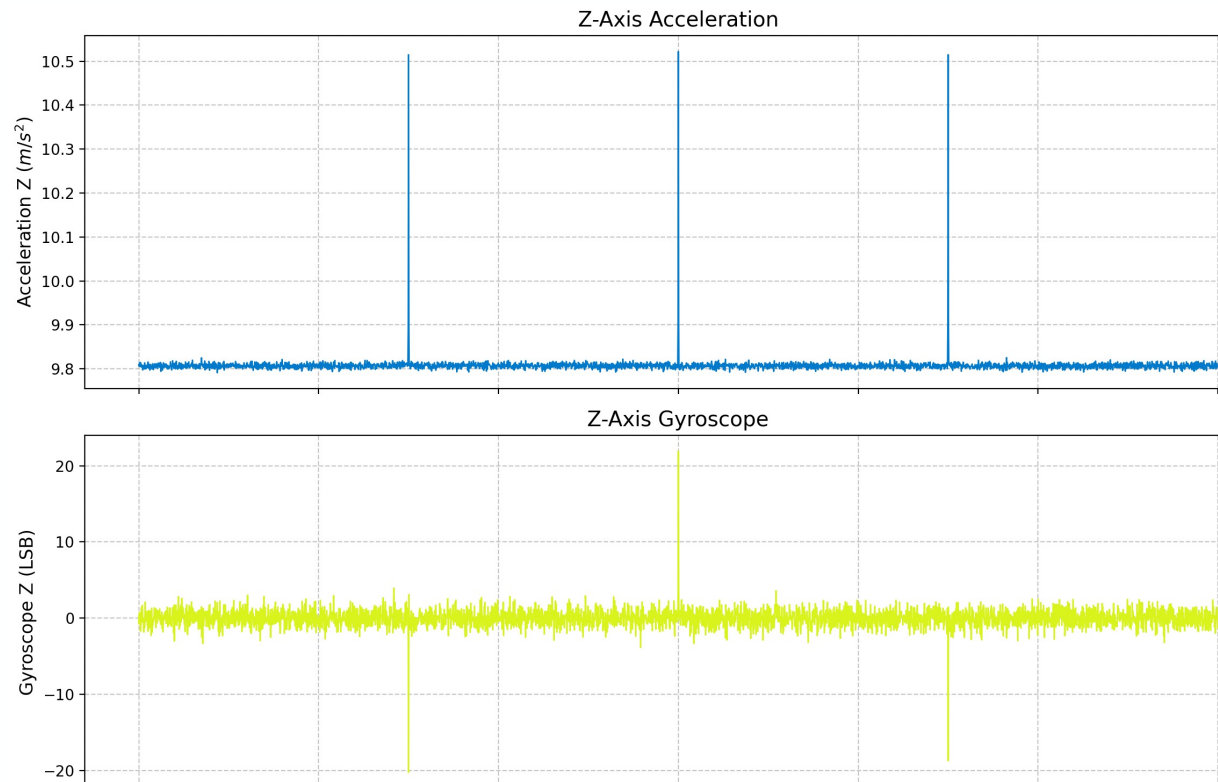
    ax2.xaxis.set_major_formatter(mdates.DateFormatter('%H:%M'))
    plt.xlabel('Time (UTC)', fontsize=12)
    plt.xticks(rotation=0)

    plt.tight_layout()
    plt.show()
```

Furthered Data Analysis of IMU

Verification Successful

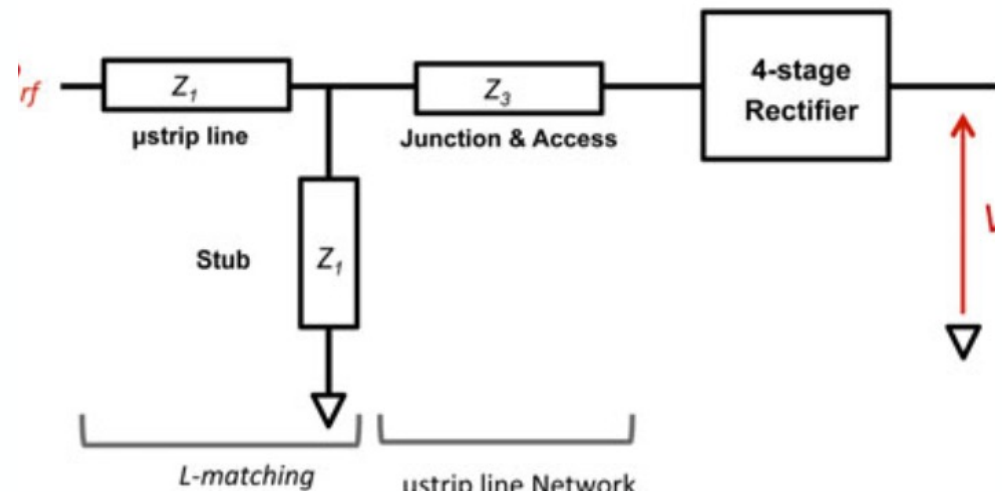
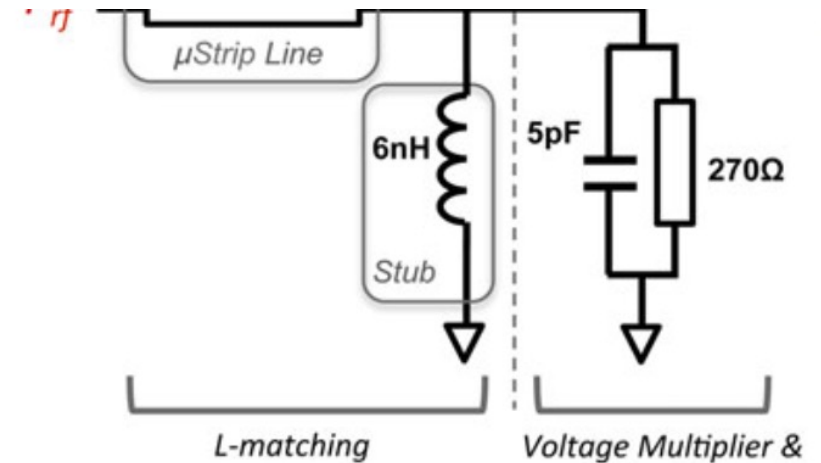
- Results seemed to match the first study
 - 9.8 m/s² acceleration amplified to around 10.5-10.8 m/s²
- Verified with magnitude and z-axis movement
 - Z-axis movement seemed to be the most substantial



Research on RF to DC Converters

915 MHz Band Research Paper

- Very little circuitry needed to run the RF to DC converter successfully
- Still need to acquire circuit components (harder than anticipated)
 - Fundamental Theory is very strong
- System and Circuit Diagrams to the right
- Works cited on last slide



DigiKey Parts Ordering

Parts List Curated – F Number Required (Procurement Office)

- **F Number** needed to subvert filling out of Form 12 repetitively
- DigiKey and Mouser Orders can be filed and filled remotely
- Cart for needed PCB components already setup:
 - <https://www.digikey.com/short/p757fqdw>
 - ~\$30 Total

☐ 1		1276-1482-1-ND CL05A475MP5NRNC Samsung Electro-Mechanics CAP CER 4.7UF 10V X5R 0402 Customer Reference	25 	Immediate	0.00900	\$0.23
☐ 2		1276-1043-1-ND CL05A104KA5NNNC Samsung Electro-Mechanics CAP CER 0.1UF 25V X5R 0402 Customer Reference	25 	Immediate	0.00200	\$0.05
☐ 3		754-2344-1-ND APTD1608F3C Kingbright INFRARED EMITTING DIODE Customer Reference	10 	Immediate	0.34700	\$3.47
Estimated Tariff: ⓘ					0.11798	\$1.18
☐ 4		1N4148WTCT-ND 1N4148WT onsemi DIODE STANDARD 75V 200MA SOD523F Customer Reference	25 	Immediate	0.06900	\$1.73
Estimated Tariff: ⓘ					0.02070	\$0.52
☐ 5		RMCF0402FT2K20CT-ND RMCF0402FT2K20 Stackpole Electronics Inc RES 2.2K OHM 1% 1/16W 0402 Customer Reference	25 	Immediate	0.00480	\$0.12
		311-4.7KJRCT-ND RC0402JR-074K7L YAGEO Customer Reference	50 			

Agriculture Components

Sent electrical components needed for study to department

- USB-A -> LiPo Charger
- JST PH 2mm 2-Pin to Female Socket Cable - 200mm



Next Week

Plans & Blockers

- Begin foundations of research paper (Overleaf)
- Ensure that agriculture study has gone successfully and there are no blockers
- Research American PCB vendors to order assembled PCB from
 - Unable to use Chinese vendors (PCBWay, JLCPCB) anymore

Works Cited

- J. G. D. P. Pinto, H. J. C. P. Oliveira, and N. B. Carvalho, "RF energy harvesting and remote powering at 900MHz and 2.4GHz," in *2015 IEEE Wireless Power Transfer Conference (WPTC)*, Boulder, CO, USA, 2015, pp. 1-3.

Thank You

Purdue Polytechnic Institute



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